
DAWN-SRW: Explicit Read-Write Operators in Learned Operation Spaces

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Abstract

Dense Transformers place computation inside large dense transformations, leaving the identity of the state transformations executed for a particular input implicit. We introduce DAWN-SRW, which formulates language-model computation as state-conditioned addressing in learned operation spaces populated by explicit read-write (RW) state-transition operators. A residual state is mapped to operation queries; each operator has an address derived from its own read and write directions; and selected rank-1 operators are composed to construct attention queries, keys, values, and a post-attention residual-state transition. At 393.8M parameters and 39.999B training tokens, DAWN-SRW retains language-model performance relative to a parameter- and token-matched dense Transformer: the C4 cross-entropies are 2.9402 and 2.9359, and six-task zero-shot means are 0.4629 and 0.4597, respectively. A descriptive 1.272B-parameter evaluation reaches 2.8359 C4 cross-entropy and a 0.4858 zero-shot mean. Mechanistically, a discovery-frozen 4,096-site Q/K-centered RW operator circuit passes suppression, four matched-control families, and exact restoration on independent validation and fully held-out indirect-object-identification inputs. The model also exhibits exact input-conditioned functional support, but the current dense implementation evaluates full operator pools and is not hardware-efficient. These results establish explicit addressed RW operators as a viable and causally analyzable computational substrate for language modeling, without claiming benchmark superiority or current sparse-execution efficiency.

1. Introduction

Transformers preserve and refine information in a residual stream, but the computational units that change that state are

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usually implicit inside dense matrices. Multi-head attention and feed-forward layers expose useful module boundaries, yet they do not report which individually identifiable state transformations were chosen for a token. This makes it difficult to separate the question *what is represented now?* from *which transformation should execute now?*

Prior work provides an important read-write view of dense feed-forward computation: a feed-forward layer can be analyzed as a collection of keys that detect input patterns and values that contribute outputs (Geva et al., 2021). That view identifies functionally meaningful directions in an existing dense layer. It does not, by itself, make each unit a globally addressable operator object whose selection is exposed and whose execution can be reused across depth.

We ask:

Can language-model computation be organized by addressing explicit read-write state-transition operators in learned operation spaces?

DAWN-SRW answers this question with two architectural choices. First, a *learned operation space* contains addresses for executable transformations rather than representations of the current token state. Second, each computational unit is an explicit rank-1 *RW operator*: it reads a scalar from the residual state and writes that scalar along a learned direction. State-conditioned addressing maps the current residual state to an operation query, compares that query with operator addresses derived from the operators’ own read and write functions, and composes the admitted operators.

The same mechanism constructs the query, key, and value states used by causal self-attention and the residual-state transition after attention. Q and K make separate selections from one shared QK pool; V and the post-attention residual transition (RST) use separate pools. The pools and addressing projections are reused across depth, while normalization and attention output projections remain layer-specific. Thus, represented state and executable operation identity are separated without claiming that the evaluated model executes read-write operations in a low-dimensional residual space.

The empirical study follows a fixed evidence order. We first compare a 393.8M parameter DAWN-SRW checkpoint with a parameter- and token-matched dense Transformer on C4

and six zero-shot tasks. We then report descriptive 1.272B results, measure exact input-conditioned operator support, and localize a discovery-only indirect object identification (IOI) circuit. The circuit is frozen before confirmatory data are opened, then tested with suppression, matched controls, and exact restoration on independent validation and fully held-out test inputs. Finally, we report three fail-closed boundaries: RAVEL localization, ARC causal validation, and 1.3B IOI localization. Efficiency is not a headline result: the current implementation scores and applies full operator pools densely.

Our contributions are:

1. **Learned operation spaces.** Language-model states address executable transformations in learned coordinate spaces rather than invoking only fixed layer-local dense maps.
2. **Explicit RW operators.** Each unit is an explicit rank-1 read-write state transition, and its address is derived from its functional read and write parameters.
3. **Attention-wide and cross-depth reuse.** Reusable, route-specific pools construct Q, K, V, and post-attention residual transitions; Q and K share a pool while retaining separate selections.
4. **Empirical and causal evidence.** We report matched 400M performance retention, descriptive 1.3B evaluation, exact functional support, and a discovery-frozen IOI circuit confirmed by validation and held-out suppression, controls, and restoration.

2. Related Work

2.1. Read-write views of Transformer computation

The Transformer applies attention and position-wise feed-forward networks to a residual stream (Vaswani et al., 2017). Geva et al. interpret Transformer feed-forward layers as key-value memories: first-layer weights detect input patterns and second-layer weights contribute output directions (Geva et al., 2021). This provides the closest conceptual precursor to our read/write factorization. DAWN-SRW changes the architectural status of that factorization. Its read and write directions directly define an executable rank-1 state transition, and its address is constructed from those same functional parameters.

Memory layers make parameter access explicit. Product Key Memory uses factorized keys for efficient exact retrieval from large learned memories (Lample et al., 2019); later work scales trainable key-value lookup to much larger parameter counts (Berges et al., 2024). These systems retrieve stored values. DAWN-SRW instead addresses oper-

ators that first read from the current state and then write a state-dependent scalar back into the representation.

2.2. Sparse addressing and fine-grained conditional computation

Sparsely-gated mixtures of experts conditionally activate a small subset of large expert networks (Shazeer et al., 2017), and Switch Transformers show that simple routing can scale this pattern (Fedus et al., 2022). PEER (parameter efficient expert retrieval) moves to much finer granularity by using product-key retrieval over more than a million small experts (He, 2024). PEER is the closest prior work at the level of atomic conditional computation: it retrieves singleton feed-forward experts. DAWN-SRW addresses explicit RW state-transition operators, derives each address from the operator’s read/write function, applies the mechanism to Q, K, V, and RST, and reuses route-specific pools across depth. Conversely, the current DAWN-SRW implementation does not provide PEER-style indexed sparse execution; it performs a dense address scan and dense, chunked RW application.

2.3. Shared computation and operation reuse across depth

Universal Transformers reuse a recurrent transformation across depth (Dehghani et al., 2019); ALBERT reduces parameters through cross-layer sharing (Lan et al., 2020). Sparse Universal Transformers combine depth sharing with sparse experts (Tan et al., 2023), and MoEUT adapts mixtures of experts to shared-layer language models (Csordás et al., 2024). MoEUT is a close architectural comparison for global reuse, but its reusable objects are expert modules inside a shared Transformer layer. DAWN-SRW reuses explicit route-specific RW pools and addressing projections while retaining layer-specific LayerNorm parameters and attention output projections. It therefore does not repeat one completely shared block.

Dynamic parameterization offers another route to input-dependent functions. HyperNetworks generate weights for another network (Ha et al., 2017), while CondConv forms example-conditioned convolutional kernels from learned experts (Yang et al., 2019). DAWN-SRW does not synthesize a dense weight matrix. It exposes a composition of fixed, identifiable rank-1 RW operators and ties their addresses to their execution directions.

2.4. Mechanistic analysis and causal intervention

Mechanistic interpretability studies model behavior through identifiable components and causal interventions. Work on IOI isolates an attention-head circuit and evaluates faithfulness, completeness, and minimality (Wang et al., 2023).

Activation patching and scalable approximations localize behavior to components (Kramár et al., 2024). RAVEL emphasizes that neuron-level representations may be entangled and evaluates methods against causal criteria (Huang et al., 2024). Our analysis treats a layer, route, and operator identity as a site, freezes localization before confirmatory evaluation, and uses direct numerator suppression, matched random circuits, and exact same-example restoration. The explicit unit helps specify interventions, but does not guarantee a human-readable interpretation.

3. DAWN-SRW

3.1. Overview

Let a token’s residual state be $x \in \mathcal{X} = \mathbb{R}^D$. For each route $a \in \{Q, K, V, \text{RST}\}$, the model maps x to an operation query $q_a(x) \in \mathcal{Z}_a$. An operation space is a learned coordinate space whose points identify executable state-transition operators rather than represented states. Operator i has address $k_{a,i} \in \mathcal{Z}_a$, and the geometry between $q_a(x)$ and $k_{a,i}$ determines whether its read-write function executes.

The conceptual transition for route a is

$$T_a(x) = x + \sum_i G_a(q_a(x), K_a(r_{a,i}, w_{a,i})) \langle x, r_{a,i} \rangle w_{a,i}. \quad (1)$$

Here K_a builds a functional address from read and write directions, and G_a performs state-conditioned addressing. Sections 3.2–3.7 give the implemented normalization, composition denominator, and route scale omitted from this summary.

3.2. Learned Operation Spaces

Representation space and operation space serve different roles:

representation space: what is currently represented,
operation space: which transformation should execute.

The evaluated model uses route width $R = 256$. One shared pair of learned probes $P_r, P_w \in \mathbb{R}^{D \times R}$ maps the functional read/write directions of every route to address features. With $\text{unit}(z) = z / \max(\|z\|_2, \epsilon)$, the implemented operator address is

$$\begin{aligned} \bar{r}_{a,i} &= \text{unit}(r_{a,i}), & \bar{w}_{a,i} &= \text{unit}(w_{a,i}), \\ u_{a,i} &= \text{unit}(\bar{r}_{a,i}^\top P_r), & v_{a,i} &= \text{unit}(\bar{w}_{a,i}^\top P_w), \\ k_{a,i} &= \text{unit}(u_{a,i} \odot v_{a,i}). \end{aligned} \quad (2)$$

This generalized coordinate-wise bilinear construction makes address identity a learned function of what the operator reads and where it writes. It is not an arbitrary expert ID. The shared probes align QK, V, and RST addresses to the same route-width coordinate convention, but the operator rows themselves belong to separate route-specific pools.

3.3. Explicit Read-Write Operators

Each operator has a normalized read direction $\bar{r}_{a,i}$ and write direction $\bar{w}_{a,i}$. Its executable function is

$$\mathcal{O}_{a,i}(x) = \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (3)$$

This is an explicit rank-1 state transition. The read direction specifies what scalar is extracted from the current state; the write direction specifies where that scalar contributes. Unlike a retrieved static value, the output depends on the current residual state even after the operator is selected.

3.4. State-Conditioned Operator Addressing

The normalized operation query is a direct projection of the route input:

$$q_a(x) = \text{unit}(x P_a + b_a). \quad (4)$$

A separate direct projection produces a token-dependent boundary:

$$\tau_a(x) = -1 + 2\sigma(x u_a + c_a), \quad (5)$$

$$\rho_{a,i}(x) = q_a(x)^\top k_{a,i}, \quad (6)$$

$$m_{a,i}(x) = \rho_{a,i}(x) - \tau_a(x). \quad (7)$$

The evaluated `linear_angular` rule uses

$$g_{a,i}(x) = \text{clip}\left(\frac{m_{a,i}(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right). \quad (8)$$

Thus exact functional support is the set $\{i : m_{a,i}(x) > 0\}$. Evaluation pruning is disabled, so the admission and execution weights are both $g_{a,i}$.

Composition normalizes by admitted mass:

$$d_a(x) = \left[\max\left(\sum_i g_{a,i}(x), 1\right) \right]^{p_a}, \quad (9)$$

$$\Delta_a(x) = \frac{\sqrt{D/L}}{d_a(x)} \sum_i g_{a,i}(x) \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i}. \quad (10)$$

The denominator powers are $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{\text{RST}} = 1.2$. The fixed output scale is $\sqrt{D/L}$, with no learned pool-strength parameter.

3.5. Attention and Residual-State Transitions

At layer ℓ , a layer-specific normalization produces $\tilde{x}_\ell = \text{LN}_\ell^{(1)}(x_\ell)$. Separate Q, K, and V operation queries and DirectTau boundaries are computed from $\tilde{x}_\ell$. Q and K address the same 4,600-row QK pool in two separate passes; V addresses a separate 14,500-row pool. Their RW compositions form full- D Q, K, and V states, which are reshaped into

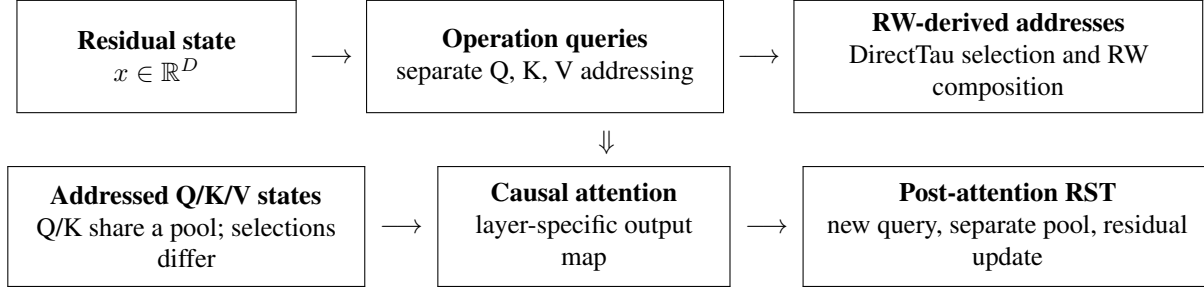


Figure 1. The evaluated DAWN-SRW block. Global route-specific operator pools and addressing parameters are reused at every depth. LayerNorm parameters and the attention output projection are layer-specific. RW execution remains in the full D -dimensional representation space.

heads and passed through standard causal scaled dot-product attention:

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M_{\text{causal}}\right) V_\ell, \quad (11)$$

$$y_\ell = x_\ell + A_\ell W_\ell^O. \quad (12)$$

After attention, a second layer-specific normalization $\tilde{y}_\ell = \text{LN}_\ell^{(2)}(y_\ell)$ is newly addressed by the RST route. The resulting composition uses its own 29,300-row pool:

$$x_{\ell+1} = y_\ell + \Delta_{\text{RST}}(\tilde{y}_\ell). \quad (13)$$

RST therefore depends on the contextualized post-attention state, not the pre-attention route query.

3.6. Global Reuse Across Depth

The QK, V, and RST read/write pools; the two RW-to-address probes; the Q/K/V query projection; the RST query projection; and their DirectTau projections are shared across all L layers. Q and K share read/write rows but retain separate query slices, tau values, gates, and outputs. The layer-specific parameters are two LayerNorms and the attention output projection W_ℓ^O . Positional and token embeddings and the final normalization are model-level parameters. Consequently, DAWN-SRW provides cross-depth operation reuse without claiming complete block sharing.

3.7. Implemented Execution and Compute Boundary

The implementation shards operator rows across the model mesh and processes them in dense chunks. The 400M configuration uses one QK chunk, one V chunk, and four RST chunks. Every query scores the full local address pool; every operator row participates in dense read and write contractions before cross-shard reduction. Exact zeros in Equation 8 therefore describe *functional* support, not an indexed sparse kernel.

The evaluated architecture also keeps RW execution in the full residual dimension D . The operation space is low-

dimensional only as an address space. A hypothetical executor can omit exactly zero RW terms without changing the mathematical output, but this conditional execution opportunity is not implemented or timed here.

4. Experimental Setup

4.1. Models and training

Table 1 summarizes the evaluated checkpoints. The primary DAWN-SRW model has 393,800,708 measured parameters, width 2,304, 18 layers, 36 heads of width 64, route width 256, and 48,400 operator rows across QK, V, and RST. It was trained for 39,999,504,384 packed C4 tokens with sequence length 512 and global training batch 1,024. Model and router dropout are zero.

4.2. Matched 400M dense control

The dense control has 393,896,960 measured parameters, width 1,280, feed-forward width 5,120, 18 layers, and 20 heads. It uses the same tokenizer, vocabulary, sequence length, training-token count, seed, batch size, mesh, learning rate, warmup ratio, weight decay, and zero dropout as the primary model. The parameter difference is 96,252, or 0.0244% of the dense model. This comparison tests performance retention under matched scale and training exposure; it is not a formal equivalence test.

4.3. Evaluation protocols

The C4 protocol evaluates 610 complete global batches from the fixed 10M-token capped packed-validation view: 19,520 sequences and 9,974,720 valid next-token targets. Cross-entropy, perplexity, and next-token accuracy use float32 scoring. The zero-shot protocol uses `lm-eval` 0.4.2 with no few-shot examples and no task limit on LAMBADA, HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and WinoGrande. The primary metric is raw accuracy for LAMBADA and WinoGrande and normalized accuracy for the other four tasks. The mean is an unweighted mean of these six metrics.

Table 1. Model and training specification. The 1.3B row is descriptive and does not form a controlled scaling curve with the 400M rows.

Model	Parameters	Tokens	D	Layers	Heads	QK / V / RST rows	Checkpoint
DAWN-SRW 400M	393,800,708	39,999,504,384	2,304	18	36	4,600 / 14,500 / 29,300	76,293
Dense Transformer 400M	393,896,960	39,999,504,384	1,280	18	20	—	76,293
DAWN-SRW 1.3B	1,272,038,404	19,999,981,568	4,096	24	32	8,534 / 26,904 / 54,352	87,193

Table 2. Packed C4 validation on the identical 9,974,720 target tokens.

Model	CE	PPL	Accuracy
DAWN-SRW 400M	2.940213	18.919873	0.424912
Dense 400M	2.935894	18.838336	0.427100
DAWN-SRW 1.3B	2.835879	17.045382	0.437099

4.4. Mechanistic localization and frozen intervention protocol

We define a site by layer, route, token position, and global operator identity. IOI discovery uses 119 independent paired-correct units. Sites are ranked by absolute discovery mean contribution importance. We choose the smallest audited prefix that reaches at least 0.70 cumulative absolute importance and 0.95 split top- k overlap. The resulting 4,096-site identity, route/layer composition, score, and intervention specification are frozen before validation.

The primary causal score is the summed-log-probability margin between the correct indirect object and the source name. Circuit-wide suppression zeros only the selected operators’ execution numerator at target positions while retaining the full production denominator. Four control families each contain 100 frozen-size circuits that exclude circuit sites: equal-count random, same-layer random, activation-matched, and route-frequency-matched. Exact restoration reinserts the same-example selected numerator after suppression, again under the production denominator. Validation uses 123 independent paired-correct units; the held-out test accessor is opened only after the validation result and unchanged specification are finalized. We report 95% bootstrap confidence intervals and two-sided paired permutation tests where specified.

5. Results

5.1. Language-model performance retention at 400M

Table 2 reports the locked C4 evaluation. DAWN-SRW differs from the dense control by +0.004318918 cross-entropy, +0.081537182 perplexity, and −0.002188432 accuracy (−0.2188 percentage point). These small measured gaps support performance retention, not superiority or formal equivalence.

Table 3 gives task-level zero-shot results with standard er-

rors. The matched 400M means are 0.462872 and 0.459652, a difference of +0.003220495 (+0.3220 percentage point). Task differences vary in sign, so the mean does not establish benchmark superiority. LAMBADA perplexities are 24.0774 ± 0.8005 for DAWN-SRW 400M, 24.0042 ± 0.8073 for dense 400M, and 17.6358 ± 0.5390 for DAWN-SRW 1.3B.

5.2. Descriptive larger-scale evaluation at 1.3B

The 1.272B checkpoint reaches 2.835879 C4 cross-entropy, 17.045382 perplexity, 0.437099 token accuracy, and a 0.485844 six-task mean. It was trained for 19.999B tokens, about half the 400M training-token count, and no completed matched dense 1.3B result is admitted to this study. These results therefore describe a larger trained DAWN-SRW model; they are neither a controlled scaling curve nor a dense-model comparison.

5.3. Input-conditioned RW operator support

Across 9,994,240 packed-C4 positions, every route has a strict subset of its pool at positive selection margin (Table 4). Mean exact support is 487.896 Q operators, 386.722 K operators, 812.388 in the shared-pool Q/K union, 1,649.994 V operators, and 2,017.703 RST operators. This is exact input-conditioned functional support under Equation 8.

The support does not make the present kernel efficient. Static forward accounting at batch 1 and sequence length 512 is 4.741260116 TFLOPs for current dense DAWN-SRW, 0.818230250 TFLOPs for dense addressing plus exact-support-only RW application, and 0.386547057 TFLOPs for the matched dense Transformer. The exact-support opportunity is 17.2577% of current DAWN-SRW accounting but still 2.1168 times the dense Transformer (Figure 2). It is a mathematical conditional-RW opportunity, not measured latency, energy, or hardware speedup.

5.4. IOI circuit localization

Discovery produces split-half rank stability 0.9139788 over 456,697 ranked sites. The smallest prefix passing both frozen discovery gates contains 4,096 sites and 0.719249 cumulative absolute importance. It spans all 18 layers and contains 1,744 Q sites, 2,350 K sites, two V sites, and no RST sites. Split top- k overlap is 0.974854 (Jaccard 0.950941). These statistics identify a Q/K-centered candidate circuit;

Table 3. Zero-shot accuracy. Parentheses contain standard errors. HS, AE, and AC denote HellaSwag, ARC-Easy, and ARC-Challenge.

Model	LAMBADA	HS	PIQA	AE	AC	WinoGrande	Mean
DAWN-SRW 400M	0.352028 (0.006654)	0.458375 (0.004972)	0.712187 (0.010563)	0.484007 (0.010255)	0.256826 (0.012767)	0.513812 (0.014047)	0.462872
Dense 400M	0.353192 (0.006659)	0.457080 (0.004971)	0.698041 (0.010712)	0.475589 (0.010248)	0.249147 (0.012639)	0.524862 (0.014035)	0.459652
DAWN-SRW 1.3B	0.395886 (0.006813)	0.510157 (0.004989)	0.723069 (0.010440)	0.509680 (0.010258)	0.268771 (0.012955)	0.507498 (0.014051)	0.485844

Table 4. Exact functional support at selection margin > 0 . Q/K union counts an operator active for Q or K at the same layer and token.

Route	Pool	Mean active	Pool fraction
Q	4,600	487.896	10.6064%
K	4,600	386.722	8.4070%
Q/K union	4,600	812.388	17.6606%
V	14,500	1,649.994	11.3793%
RST	29,300	2,017.703	6.8864%

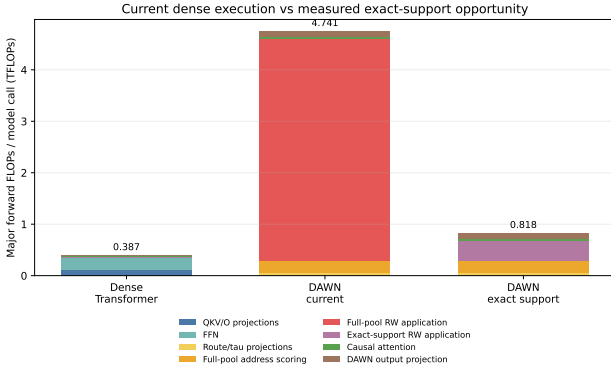


Figure 2. Static backbone accounting at batch 1 and sequence length 512. Address scoring remains dense in the exact-support estimate. Current execution also applies all full-dimensional RW rows. No production latency claim is made.

localization alone is not a causal claim.

5.5. Validation and held-out causal intervention

Table 5 shows confirmatory results. On validation, suppression moves the mean margin from 3.465456 to -4.399331 , a drop of 7.864787 (95% CI [7.490080, 8.241263]), and accuracy falls from 1.000000 to 0.008130. The frozen effect exceeds every replicate in all four 100-circuit control families. The most competitive control is activation-matched, with mean drop 5.400490; the frozen-minus-control difference is 2.464297 (95% CI [2.076200, 2.832697]), paired permutation $p = 0.00049975$. Exact restoration returns accuracy to 0.991870 and recovers 0.999661 of the suppression effect (95% CI [0.998655, 1.000584]).

The fully held-out test reuses the unchanged site identity, scoring rule, intervention, controls, seeds, and thresholds. Suppression moves margin from 3.254720 to -4.155613 , a drop of 7.410332 (95% CI [7.016485, 7.818472]), while accuracy falls to 0.008772. Again the frozen effect exceeds

every replicate in all four control families. The activation-matched mean drop is 5.472579; the frozen-minus-control difference is 1.937754 (95% CI [1.564818, 2.319706], $p = 0.00049975$). Restoration recovers 0.998406 of the effect (95% CI [0.997324, 0.999385]) and restores accuracy to 1.000000.

Together, suppression, matched-control separation, and exact restoration support the claim that this discovery-frozen Q/K-centered operator circuit is causally necessary for the evaluated IOI behavior on independent validation and fully held-out inputs. “Necessary” is intervention-specific: it does not mean the sites form a unique or human-readable algorithm.

5.6. Negative and fail-closed results

The same workflow produces informative negative boundaries (Table 6). RAVEL passes the behavioral gate but fails variable-localization stability, so no circuit is frozen and no confirmatory split is opened. ARC discovery freezes a stable 4,096-site Q/K circuit, but validation suppression has the wrong sign and fails control separation; restoration passes, and test remains closed. For 1.3B IOI, behavioral eligibility and rank stability pass, but the largest allowed 4,096-site prefix reaches only 0.635144 cumulative importance, below the preregistered 0.70 threshold. No circuit is frozen and no validation or test is opened.

6. Discussion

6.1. What the results establish

The evaluated model shows that a language model can construct attention and post-attention residual updates by state-conditionally addressing explicit RW operators. The matched 400M comparison shows performance retention under the measured protocols. The support profile shows that the gate has exact input-conditioned zeros, and the IOI study shows that operator sites can form a discovery-frozen circuit whose suppression and restoration causally control behavior on held-out data.

These claims are deliberately narrower than a theory of language reasoning as discrete programs. The causal result is strong for the specified IOI task and intervention, not universal across tasks or scales. RAVEL, ARC, and 1.3B IOI demonstrate that explicit units do not make successful localization automatic.

Table 5. Frozen IOI circuit results. Accuracy arrows show intact to suppressed. Recovery is the fraction of the suppression effect recovered by exact same-example numerator restoration.

Split	Units	Intact margin	Suppressed margin	Drop (95% CI)	Accuracy	Recovery / restored acc.
Validation	123	3.465456	-4.399331	7.864787 [7.490080, 8.241263]	1.000000 → 0.008130	0.999661 / 0.991870
Held-out test	114	3.254720	-4.155613	7.410332 [7.016485, 7.818472]	1.000000 → 0.008772	0.998406 / 1.000000

Table 6. Fail-closed mechanistic results. Later phases were not opened after a failed preregistered gate.

Study	Passed	Failed boundary	Decision
RAVEL 400M	Behavioral eligibility	Variable stability: 0.628762 / 0.676243 / 0.660622	No frozen circuit; no validation, transfer, or test
ARC 400M	Stable discovery; 4,096 Q/K sites; restoration	Suppression effect wrong sign; control separation failed	Reject causal replication; held-out test closed
IOI 1.3B	Behavioral eligibility; rank stability 0.907188	4,096-site cumulative importance 0.635144 < 0.70	No frozen circuit; validation and test closed

6.2. Operation spaces as a computational abstraction

An operation space separates represented state from executable operation identity. A query describes what transformation is currently relevant; an address identifies a reusable transformation; and an RW operator specifies the executed function. Because the address is derived from read/write parameters, selection geometry and execution function are coupled rather than assigned independent labels.

This coupling is an architectural constraint, not proof of semantic interpretability. An operator may be explicit, observable, and causally testable while participating in distributed or polysemantic computation. The IOI circuit’s 4,096 sites across all layers is consistent with this distributed view.

6.3. Toward Low-Dimensional Operation-Space Execution

The evaluated model uses low-dimensional addresses but full- D RW execution. A separate repository implementation is an *implemented but not yet fully evaluated low-dimensional extension*. It associates each operation-space address with $D \leftrightarrow R$ interfaces, selects spaces independently for Q, K, V, and RST, and performs local RW composition before writing back:

$$h_m = P_m^\top x, \quad (14)$$

$$z_{a,m} = \sum_i g_{a,m,i}(h_m) \langle h_m, r_{a,m,i} \rangle w_{a,m,i}, \quad (15)$$

$$\Delta x_a = \sum_m \alpha_{a,m}(x) B_m z_{a,m}. \quad (16)$$

Its local operator remains globally rank-1 because $B_m w_i r_i^\top P_m^\top = (B_m w_i)(P_m r_i)^\top$. Compilation, symbolic parameter counts, reduced JIT forward/backward, and gradient contracts have been checked, and limited early-training observations exist. There is no completed controlled language-model evaluation or realized efficiency measurement. The current reference path executes all spaces

densely; candidate-restricted dispatch and an accelerator-native kernel remain future work. Appendix G records the boundary in more detail.

6.4. Limitations

The current implementation evaluates full address and operator pools densely. Exact functional support does not imply current sparse execution, and even the exact-support accounting remains 2.1168 times the matched dense Transformer’s backbone accounting. No production latency, energy, or hardware advantage is measured.

The central causal evidence is confined to IOI and does not automatically generalize to every task. RAVEL, ARC, and the selected 1.3B IOI replication provide valid negative boundaries. The 400M and 1.3B checkpoints have different parameter and training-token budgets, so they do not define a controlled scaling curve. Operator-level causal units need not map one-to-one to human-readable concepts. The low-dimensional extension has not received a controlled evaluation. Finally, the present work does not establish language reasoning as universally discrete operator programs.

7. Conclusion

DAWN-SRW formulates language-model computation as state-conditioned addressing in learned operation spaces populated by explicit RW state-transition operators. The architecture separates represented state from operation identity, derives addresses from read/write function, and reuses route-specific operator pools to construct Q, K, V, and post-attention residual transitions across depth. It retains measured language-model performance relative to a matched 400M dense control. A discovery-frozen Q/K-centered operator circuit is causally necessary under suppression and recoverable under exact restoration on independent validation and fully held-out IOI inputs. The model also shows exact input-conditioned functional support. These findings

support addressed RW operators as a viable and causally analyzable computational substrate, but do not establish benchmark superiority, universal operator programs, or current hardware efficiency.

A. Exact Addressing and Composition Equations

This appendix restates the production computation as a compact reference. Raw read and write parameters are normalized in float32 before being cast for dense contractions:

$$\bar{r}_i = \frac{r_i}{\max(\|r_i\|_2, \epsilon)}, \quad \bar{w}_i = \frac{w_i}{\max(\|w_i\|_2, \epsilon)}.$$

One pair of model-level probes is shared across QK, V, and RST:

$$\begin{aligned} u_i &= \text{unit}(\bar{r}_i^\top P_r), & v_i &= \text{unit}(\bar{w}_i^\top P_w), \\ k_i &= \text{unit}(u_i \odot v_i). \end{aligned}$$

The query and threshold use separate direct projections:

$$\begin{aligned} \hat{q}_a &= \text{unit}(xP_a + b_a), \\ \tau_a &= -1 + 2\sigma(xu_a + c_a), \quad \rho_i = \hat{q}_a^\top k_i. \end{aligned}$$

For `linear_angular` composition:

$$g_i = e_i = \text{clip}\left(\frac{\rho_i - \tau_a}{\max(1 - \tau_a, 10^{-4})}, 0, 1\right)$$

because evaluation pruning is disabled. Padding rows are masked to zero. Let $M_a = \sum_i g_i$. The production route result before causal attention or residual addition is

$$\Delta_a(x) = \sqrt{\frac{D}{L}} \frac{\sum_i e_i \langle x, \bar{r}_i \rangle \bar{w}_i}{\max(M_a, 1)^{p_a}},$$

with $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{RST} = 1.2$. Operator rows are sharded over the model axis; local numerators and admission mass are reduced across shards. Chunking changes the contraction schedule, not the equation.

B. Complete Model and Training Settings

The 400M DAWN-SRW checkpoint is the primary performance and mechanistic model. The dense 400M checkpoint is only a matched performance control. The 1.3B DAWN-SRW checkpoint is used for descriptive performance and one fail-closed IOI localization replication. The still-incomplete dense 1.3B run is excluded.

C. Benchmark Metrics and Protocol Details

C4. All reported C4 rows use the same packed validation artifact and `packed_c4_validation_v1` protocol. The

capped view contains 19,531 sequences. Exactly 19,520 sequences form 610 complete global evaluation batches, producing 9,974,720 valid next-token targets; 11 incomplete-tail sequences are excluded before scoring.

Zero-shot. The protocol is `mamba_table3_zero_shot_v1`, using `stock_lm_eval==0.4.2`, global batch 32, maximum length 512, length buckets 64/128/256/512, and zero few-shot examples. Evaluated split sizes are 5,153 for LAMBADA, 10,042 for HellaSwag, 1,838 for PIQA, 2,376 for ARC-Easy, 1,172 for ARC-Challenge, and 1,267 for WinoGrande. Task, dataset, tokenizer, and software contracts match exactly between the two 400M models.

D. IOI Localization and Intervention Details

D.1. Frozen discovery

Discovery admits 119 paired-correct independent units and ranks 456,697 sites. Split-half rank stability is 0.913978765. The frozen selection rule is the smallest prefix satisfying cumulative absolute importance at least 0.70 and split top- k overlap at least 0.95. The selected $k = 4096$ prefix has:

- cumulative absolute importance 0.719248713;
- 3,993 shared split sites, overlap 0.974853516, and Jaccard 0.950940700;
- route counts Q/K/V/RST = 1744/2350/2/0;
- coverage of every layer from 0 through 17.

Validation and test do not recompute or alter this selection.

D.2. Suppression, controls, and restoration

For a correct candidate c and source candidate s , the primary score is

$$\mathcal{M} = \sum_{t \in c} \log p(t) - \sum_{t \in s} \log p(t).$$

Suppression zeros the frozen sites' execution numerator at specified token positions. It leaves admission mass and the production denominator unchanged. Controls use the same 4,096-site count and exclude all circuit sites. Activation matching conditions on layer, route, and the discovery mean pre-scale operator-output-norm quantile. Restoration executes a distinct intervention: after suppression, it adds back the exact selected same-example numerator under the unchanged denominator.

On validation, restoration returns the margin to 3.462791, leaving residual drop 0.002665. On held-out test, it returns the margin to 3.242905, leaving residual drop 0.011815.

Table 7. Detailed settings from the registered training configurations and restored checkpoint manifests.

Setting	DAWN-SRW 400M	Dense 400M	DAWN-SRW 1.3B
Vocabulary	30,522	30,522	30,522
Residual width	2,304	1,280	4,096
Layers	18	18	24
Heads / head width	36 / 64	20 / 64	32 / 128
Maximum sequence length	512	512	512
Feed-forward width	–	5,120	–
Operation-route width	256	–	256
QK / V / RST pool	4,600 / 14,500 / 29,300	–	8,534 / 26,904 / 54,352
QK / V / RST chunks	1 / 1 / 4	–	1 / 2 / 8
QK / V / RST denominator power	0.5 / 1.0 / 1.2	–	0.5 / 1.0 / 1.2
Initial active targets	0.05 / 0.03 / 0.01	–	0.05 / 0.03 / 0.01
Training batch	1,024	1,024	448
Mesh (data $\times$ model)	16×2	16×2	16×2
Learning rate	3×10^{-4}	3×10^{-4}	2×10^{-4}
Warmup ratio	0.08	0.08	0.06
Weight decay	0.1	0.1	0.1
Tau learning-rate multiplier	0.001	–	0.001
Dropout / router dropout	0 / 0	0 / –	0 / 0
Gradient checkpointing	yes	yes	yes
Seed	1	1	1

Table 8. Control effects and frozen-minus-control separation. Each control family contains 100 replicates. All paired permutation p -values are 0.0004997501 with 2,000 permutations.

Control	Validation drop	Frozen minus control (95% CI)	Test drop	Frozen minus control (95% CI)
Equal-count	0.070886	7.793902 [7.437851, 8.161572]	0.080276	7.330057 [6.922408, 7.716879]
Same-layer	0.039389	7.825399 [7.440629, 8.180288]	0.041073	7.369260 [6.977447, 7.754539]
Activation-matched	5.400490	2.464297 [2.076200, 2.832697]	5.472579	1.937754 [1.564818, 2.319706]
Route-frequency	0.219162	7.645626 [7.289512, 8.031625]	0.242665	7.167668 [6.779836, 7.545537]

The respective recovery fractions are 0.999661155 (95% CI [0.998654544, 1.000584125]) and 0.998405645 (95% CI [0.997323737, 0.999385266]).

E. Negative Mechanistic Results

RAVEL. Discovery uses 142 independent units. Rank stability is 0.628762 for Continent, 0.676243 for Country, and 0.660622 for Language, below the preregistered requirement. No prefix passes the complete discovery rule, no variable circuit is frozen, and validation, transfer, and test are not opened.

ARC. Discovery stability is 0.89386 and a 4,096-site Q/K-only circuit is frozen. On 51 validation units, intact margin is 1.735534 and suppressed margin is 2.819139, giving a frozen margin drop of -1.083605 (95% CI $[-1.472180, -0.690348]$). The effect has the wrong sign and fails control separation. Restoration recovery is 1.004083 (95% CI [0.994587, 1.013652]), but restoration alone does not pass the conjunction. Held-out test remains closed.

1.3B IOI. The scale replication admits 125 independent units and passes rank stability at 0.907187749. The largest

allowed 4,096-site prefix reaches only 0.635144 cumulative absolute importance, below the 0.70 gate. No circuit is frozen; validation and test remain closed.

F. Compute Accounting

The support profile evaluates 9,994,240 packed-C4 positions and observes address-stage margins before RW application. It stores aggregate statistics, not per-token operator identities or vectors. Static accounting uses batch 1, sequence length 512, and 1 MAC = 2 FLOPs. It counts major matmul/einsum operations and omits softmax, normalization, gate elementwise arithmetic, embedding lookup, residual additions, and communication collectives.

The exact-support estimate is mathematically output-equivalent if the executor can skip all and only zero-weight RW terms. It does not include indexed address retrieval and does not demonstrate kernel parity. Instrumented profiling time is excluded from latency claims because histogram collection is part of the measured path.

Table 9. Backbone static accounting. “Exact support” retains full dense address scoring and replaces only full-pool RW application with the measured positive-margin support.

Component	Current dense DAWN-SRW	Dense address + exact RW	Dense Transformer
Address / route / tau projections	0.293742	0.293742	—
RW application	4.306187	0.383157	—
Causal attention	0.043487	0.043487	0.024159
Dense output / QKV / FFN maps	0.097845	0.097845	0.362388
Total TFLOPs	4.741260	0.818230	0.386547

G. Low-Dimensional Operation-Space Extension

The low-dimensional extension uses one shared atlas of $M = 8$ operation-space addresses, shared $D \rightarrow R$ read interfaces P_m , and shared $R \rightarrow D$ write interfaces B_m . Q, K, V, and RST have independent query projections and select two atlas entries independently. Route-specific tau maps and operator banks remain separate. For local operator i in space m ,

$$\begin{aligned}\mathcal{O}_{a,m,i}^{\text{global}}(x) &= B_m w_{a,m,i} r_{a,m,i}^\top P_m^\top x \\ &= (B_m w_{a,m,i})(P_m r_{a,m,i})^\top x,\end{aligned}$$

so the induced global transformation remains rank-1.

The 40M reference has 40,341,088 parameters, $D = 432$, $R = 128$, 12 layers, six heads, and top-2 space selection. Repository-local compilation, abstract-tree and symbolic parameter counting, shared-key and independent route selection, legacy-checkpoint rejection, reduced JIT forward/backward, selector metrics, and gradient contracts have passed. Some early-training observations are recorded, but the exact deployment and complete controlled language-model evaluation are absent. No latency, throughput, energy, or hardware-efficiency result exists. The reference path executes every space densely; future work requires candidate-restricted dispatch and an accelerator-native execution kernel.

H. Reproducibility Identities

The machine-readable checkpoint registry is `configs/paper-checkpoint-registry.yaml`. Exact evaluation commits, software versions, data hashes, result roots, and complete artifact hashes are recorded in the repository experiment ledger. Historical experiment branches are evidence identities only; the present source is maintained on the canonical integration branch.

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Table 10. Primary immutable identities. Branch names are omitted because they move; exact checkpoint steps, hashes, and artifact locations are the binding identities.

Evidence	Exact identity	Repository source
DAWN-SRW 400M	model spatial-r1-v4.1.7.2; step 76,293; checkpoint identity a7ce8afc...f91c17; config hash 08733ae4...5039b	configs/train_config_v4172_400M_c4_40B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
Dense 400M	model baseline-JAX; step 76,293; config hash c3ce5d72...82fc25	configs/train_config_baseline_tpu_400M_c4_40B_v4_64_tp2.yaml
DAWN-SRW 1.3B	model spatial-r1-v4.1.7.2; step 87,193; checkpoint identity 2e38b41f...bf5f9; config hash 1e877263...4a5222	configs/train_config_v4172_1p3B_c4_20B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
IOI circuit	parsed frozen-spec hash 6ec26618...418c; site hash 573cfff8...0200; selected-row hash 1439f6f3...fe0c1	configs/paper_ioi_frozen_circuit_v4172_400m.yaml
Compute support	support hash b1eac5f5...bdc2; accounting hash d7ad6115...984e	assets/paper/v4172_400m_compute_support_step76293/

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